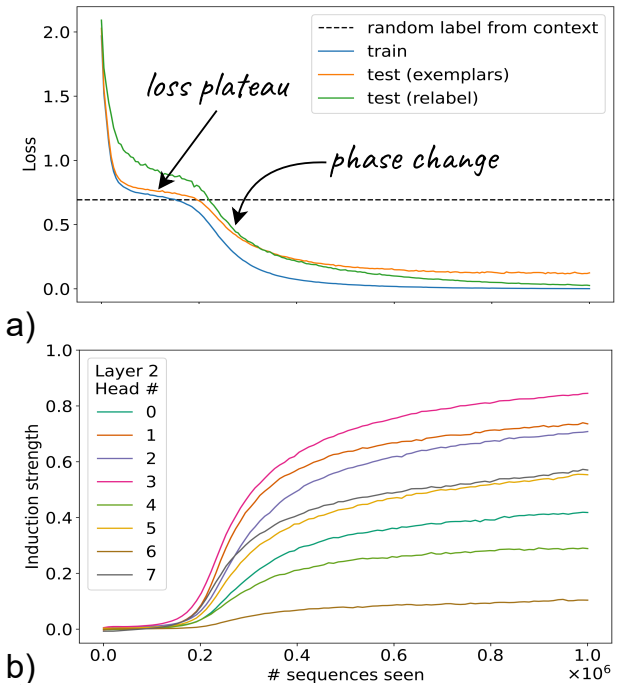

Figure 3. **a)** Train and test loss curves. Transformers exhibit strong generalization to unseen classes (orange) and label pairs (green). The loss dynamics reveal a plateau (which may be indicative of a saddle point), where the model is randomly guessing between the two labels present in context (so it has 50% accuracy, instead of the chance level of 20% when there are $L = 5$ labels). Then, there's a phase change in the loss which corresponds to the formation of induction circuits, reproducing the finding of Olsson et al. (2022). **b)** Induction head strength for each Layer 2 head plotted over time. Induction head strength is defined as the attention weight given to the correct label token minus that to the incorrect token. All heads appear to have some induction-like behavior, with Head 3 being the strongest and emerging first.

namics (see Appendix D.2 for further discussion).

3. Induction head emergence and diversity

We first establish that training transformers as described in Section 2.1 leads to ICL abilities and induction head emergence. Figure 3a shows loss dynamics. Initially, the transformer reaches a plateau (perhaps indicative of a saddle point) where its accuracy is ≈50% (a loss of $\log 2$). This corresponds to the network having learned to randomly select from the two labels in context,[8] rather than the full set of all possible labels ($L = 5$). Then, a sudden phase change in the loss leads to near-0 loss and near-perfect accuracy on the task (99.99%). At this point, we also observe strong generalization to unseen exemplars and unseen label pairs, indicating the transformer has learnt a general ICL mechanism and is not simply memorizing the training sequences.

[8]The model tends to place equal weight on the two labels, rather than randomly putting high weight on one.

We find that the phase change in the loss corresponds to the formation of induction circuits. We measure the "induction strength" of a head, a common progress measure, as a difference in attention weights from the query token: [attention to the correct in-context label token] - [attention to the incorrect in-context label token]. For example, on the sequence "A 0 B 1 A", this would be the difference [attention weight from 5th token to 2nd token] - [attention weight from 5th token to 4th token]. We see that the emergence of induction heads corresponds to the phase change in the loss.

To verify that Layer 2 induction heads instead of Layer 1 heads are primarily contributing to task performance, we use our artificial optogenetics framework to ablate the residual connection between Layer 1 heads and the output. We do this by setting Layer 2 activations (attention patterns and values vectors) to those from an un-ablated forward pass (keeping induction circuits intact) and then ablating Layer 1 head outputs. Such an ablation leads to a negligible change in loss and accuracy (which drops by 0.01% to 99.98%), indicating that Layer 1 heads are not responsible for task performance. When we apply this ablation across checkpoints, we see that earlier in training, Layer 1 heads are contributing to minimizing the loss (perhaps through the use of skip-trigrams; Elhage et al., 2021), but no longer towards the end (Appendix Figure 12).

3.1. The additive nature of induction heads

One of the notable features of Figure 3b is the fact that many heads become strong induction heads. What purpose do all these heads serve? Are they necessary to solve the task?

To answer these questions, we consider two types of ablations of the fully trained network (Figure 4a). The first type of ablation is similar to prior mechanistic work (Olsson et al., 2022), where we ablate a given head and observe the effect on network performance (triangles, Figure 4a). However, we found that this type of ablation was not very informative, and even potentially misleading, as ablating any head except the strongest leads to an almost unobservable decrease in task performance. Next, we considered an ablation inspired by work on head pruning (Michel et al., 2019), where all Layer 2 heads are turned off and only one head is turned on (circles, Figure 4a). This ablation allows us to identify the positive contribution of each head to the task. [9] As seen in Figure 4a, the latter ablation (circles) reveals a clearer picture of the importance of induction heads, showing the correlation between induction head strength (which

[9]There are some intuitive connections to the commonly used *logit lens* (nostalgebraist, 2020). However, we emphasize that our method relies on causal ablations only and takes into account LayerNorm, which is often a thorn for interpretability research. We measure task performance with the ablation applied, instead of a change in logits.

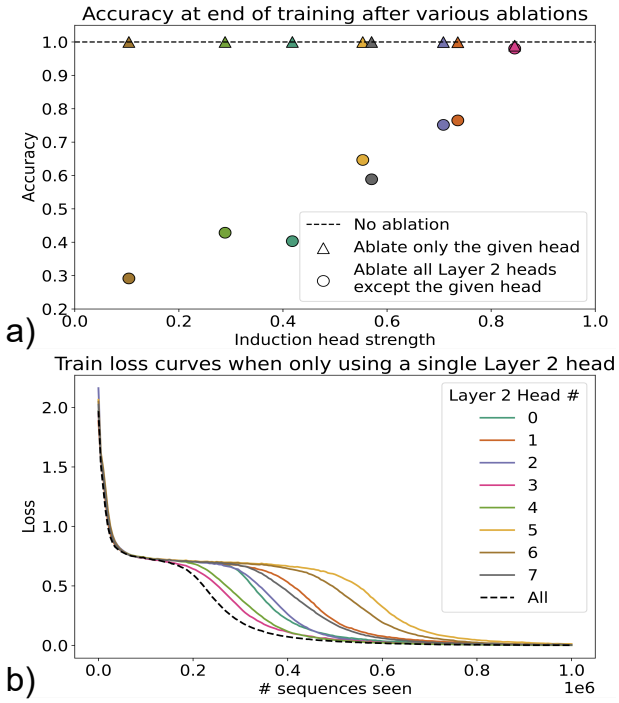

Figure 4. **a)** Effect of various ablations on accuracy (effects on loss are shown in Appendix Figure 13). Ablating any single head (triangles) leads to virtually no decrease in task performance, with the exception of Head 3, which leads to a 1% decrease. Ablating all but a specific head (circles) isolates how useful that specific head is, which correlates well to the induction strength (x-axis), the metric from Figure 3b. Importantly, ablating Head 3 (pink triangle) performs very similar to ablating everything except Head 3 (pink circle), which indicates the other heads function additively, and together can make up for the deletion of Head 3. **b)** Training loss curves when training from scratch with only a single head from Layer 2 active (and the rest ablated). Black dotted line is the loss profile from the training run in Figure 3. Colors chosen to match Figure 3b. Each Layer 2 head on its own can learn to solve the task, though the timing of the phase change shifts and learning is slower.

is calculated using attention scores), and task performance.

We see that despite Head 3 being able to mostly solve the task on its own (achieving an accuracy of 98%), other single heads can still achieve strong performance. Furthermore, heads seem to have an additive effect, most clearly demonstrated[10] by comparing the pink triangle (which corresponds to ablating Head 3) and pink circle (which corresponds to only keeping Head 3). Both achieve around 98% accuracy, indicating that the other, weaker heads together can make up for the loss of Head 3. These results echo similar findings in head pruning (Michel et al., 2019) and layer-wise redundancy of language models (McGrath et al., 2023).[11]

[10]We provide further evidence in Appendix G Figure 14.

[11]Such redundancy is also a common observation in neuroscience (Hennig et al., 2018).

Our analysis builds on this work and cautions against the use of purely knock-out ablations in mechanistic analyses, which may overlook redundancies and incorrectly dismiss network pieces as not implementing a certain function.

3.2. Networks use additional capacity for faster training, even if it is not necessary

Given the additive qualities observed above, a natural question is—does the network need multiple induction heads to *learn* to solve the task? We know that we can ablate many heads and not hurt performance *at the end of training*, but perhaps these heads play a crucial role *during training*? To answer these question, we use our artificial optogenetics framework to ablate all Layer 2 heads but one *throughout training*. We do this for each Layer 2 head and show train loss curves in Figure 4b (all networks trained from the same random initialization). We find that final networks attain the same near-perfect train performance as the network trained with 8 heads in Layer 2 (black dotted line in Figure 4b, blue line in Figure 3), but with slightly slower dynamics (delayed phase change timing). Thus, it seems the network may make use of the additional capacity during training to learn faster, even though it is not strictly necessary for the task.

These experiments connect to notions of lottery tickets[12] (Frankle & Carbin, 2018; Nanda, 2023). Specifically, we note that different Layer 2 heads exhibit phase changes at different times (but all eventually learn to solve the task). Head 3 is the "quickest to learn" in Figure 4b, which may also be the reason it becomes the strongest when training with all heads (Figure 3b)—perhaps heads are racing to minimize the loss, similar to the mechanism in Saxe et al. (2022). Once Head 3 emerges, though, the ordering of phase changes in Figure 4b does not match emergence in the full network (Figure 3b)—e.g., Head 1 is the second to emerge and second strongest, but the third slowest when trained on its own. Additive interactions could be causing Head 1 to be learned sooner, due to the presence of Head 3 (we further explore such interactions in Appendix H).

3.3. Previous token heads influence induction heads in a many-to-many fashion

Prior work (Olsson et al., 2022) has stressed the two-layer nature of induction circuits, where induction heads in later layers rely on the output of previous token heads in earlier layers to attend to the correct token. By inspecting attention patterns, we identify three previous token heads in Layer 1: Heads 1, 2, and 5. Through a series of ablation analyses, we find that each of these heads is enough to elicit above-chance accuracy in at least one of the strongest induction heads in Layer 2 (see Appendix Figure 11b). These results indicate

[12]Lottery tickets typically refer to sub-networks sufficient to solve the task determined largely by initialization.

a many-to-many wiring between previous token heads and induction heads, with previous token heads operating redundantly (just as induction heads operate redundantly to solve the task). Full details are provided in Appendix E.

4. Three interacting subcircuits give rise to the phase change in induction head formation

We now turn our attention to the dynamics of induction circuit formation. Prior work (Jermyn & Shlegeris, 2022) indicates that reverse-S phase changes in the loss are often due to multiple interacting components. Below, we delineate the necessary computations that comprise an induction circuit, then study their formation dynamics in isolation using our artificial optogenetics framework.

4.1. Terminology for this section

For clarity, we isolate and define the 5 key computations involved in an induction circuit, as illustrated in Figure 5c. In Section 4.2, we'll show that these 5 computations can be grouped into the 3 primary interacting subcircuits.

Step 1 (PT-attend) Layer 1 head attends to previous token.

Step 2 (PT-copy) Layer 1 head copies previous token value into the current token's residual stream.

Step 3 (Routing Q/K/V) Layer 2 head reads Q/K/V[13] from the correct subspaces of the residual stream: Q from the residual, K from the output of the Layer 1 head, and V from the residual or from other Layer 1 heads.[14]

Step 4 (IH-Match) Layer 2 head matches Q to "same" K.

Step 5 (IH-copy) Layer 2 copies the value to the output.

Steps 1 and 2 comprise what's canonically known as a previous token head, while Steps 3-5 form the induction head. The "match operation" (Figure 1a, blue) consists of Steps 1, 2, 3qk, 4. The "copying operation" (Figure 1a, green) consists of Steps 3v, 5.

4.2. *Clamping* computations to understand the causal effects of subcircuits on dynamics

Now that we've defined these computations, we ask: how do the learning dynamics of each of these computations causally influence the learning dynamics of the full model? Can we explain seemingly discontinuous qualitative phase changes, such as the induction circuit formation, in terms of subcircuits with smoother, exponential learning dynamics?

To motivate our analysis, we extend the toy model from Jermyn & Shlegeris (2022) to the case of three interacting vectors.[15] Specifically, we have three vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ that are learnt using a simple mean-squared error loss, corresponding to learning their values via a tensor product:

$$\mathcal{L}(\mathbf{a}, \mathbf{b}, \mathbf{c}) = \frac{1}{2} \sum_{i,j,k} \left(a_i^{true} b_j^{true} c_k^{true} - a_i b_j c_k \right)^2$$

The interaction between the evolution of $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ gives rise to a phase change in dynamics (black, Figure 5a) following a loss plateau caused by a saddle point (see proof in Appendix F). If we clamp $\mathbf{c}$ to its final value, learning is faster (blue, Figure 5a), but the co-evolution of $\mathbf{a}$ and $\mathbf{b}$ still results in a phase change. If we clamp $\mathbf{b}$ *and* $\mathbf{c}$ (purple, Figure 5a), we isolate a smooth exponential loss curves corresponding to the formation dynamics of $\mathbf{a}$. This toy model gives us the intuition that phase changes may be caused by two or more subcircuits that, when learned on their own, evolve exponentially, but when co-evolving, induce phase changes. Furthermore, these subcircuits could be isolated by clamping all the other interacting components throughout training. Notably, it's not clear how to use existing interpretability techniques, such as progress measures (Nanda et al., 2023; Reddy, 2023), to uncover these subcircuits (Appendix D.2).

We apply these insights from the toy model to understanding the key underlying subcircuits in an induction circuit. Specifically, we aim to isolate subcircuits where each individual subcircuit does not exhibit a phase change in its learning dynamics, like variables $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ in the toy model. We restrict ourselves to considering the induction circuit with induction head Layer 2 Head 3, a minimal setting where we observe a phase change in the loss dynamics (see Figure 4b). In Section 5, we will show how understanding these subcircuits can help us understand the data-dependence of phase change timing, affirming the importance of discovering smoothly-learned underlying subcircuits.

To isolate these subcircuits, we iteratively clamp the computations delineated in Section 4.1 at the activation level,[16] using our artificial optogenetics framework. Results are shown in Figure 5b. We start by clamping Step 1 (PT-attend), the previous token head attention pattern (an oft-used progress measure for IH formation). We train a network where one[17] of the Layer 1 heads is clamped from the start of training

[13] Q/K/V are the query, key, and value in the attention computation of a transformer (Vaswani et al., 2017).

[14] Unlike Olsson et al. (2022), we observe non-trivial V-composition in our networks. See Appendix E for details.

[15] Our tensor product formalism for analyzing sub-circuits could be extended beyond 3. We chose 3 as we ended up finding 3 primary subcircuits contributing to induction circuit formation. We also assume that $\mathbf{a}^{true}, \mathbf{b}^{true}, \mathbf{c}^{true} \neq \mathbf{0}$.

[16] Specifics of our implementation are provided in Appendix D.1.2 and why fixing weights to those at the end of training is not sufficient (see Appendix D.1.1).

[17] We also conducted experiments where multiple previous token heads were provided and observed similar loss dynamics. See Appendix D.1.2.